# Rupesh Devarapalli

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2028 | 7.50 |
| NARAYANA JR COLLEGE (08036) | Class XII (Board of Intermediate Education, Andhra Pradesh) | 2024 | 96.60% |
| DR S R K SCHOOL | Class X (Board of Secondary Education, Andhra Pradesh) | 2022 | 89.33% |

## RELEVANT COURSES

**Software Core:** Software Development Fundamentals, Data Structures, Algorithms, Database Management Systems.

**Systems Core:** Operating Systems, Computer Architecture, Compilers.

**Mathematics Core:** Discrete Mathematics for comp. sci., Probability & Random Variables, Linear Algebra, Theory of Computation.

## SKILLS

**Languages:** C++, C, Java, Python, Flex, RISC-V, SQL, GDscript.

**Softwares:** PostgreSQL, Godot, Blender.

## PROJECTS

**Extending xv6 operating system** | C, xv6 | Course Project

- Architected and implemented core resource management subsystems, including CPU scheduling, memory management, and storage processing for xv6-riscv (MIT PDOS) operating system.
- Upgraded the xv6 kernel by engineering a custom multi-level feedback queue (MLFQ) CPU scheduler to to balance system resources between interactive and compute-heavy workloads.
- Implemented a virtual memory paging system that gracefully handles memory exhaustion via disk-backed swapping.
- Developed a simulated software RAID storage layer (RAID 0, 1, 5) with FCFS and SSTF disk scheduling policies, integrating I/O priority directly with the CPU scheduler.

**Arbitrary-Precision Arithmetic Library** | Java | Course Project

- Developed a custom Java library (AInteger, AFloat) from scratch to perform arbitrary-precision arithmetic without relying on native BigInteger or BigDecimal classes.
- Implemented core mathematical operations (+, -, *, /) for infinitely large integers and floats through complex, manual algorithmic string manipulation.
- Designed a Java-based command-line interface (MyInfArith) to seamlessly parse user input and evaluate complex arithmetic expressions directly in the terminal.
- Applied strict Object-Oriented Programming (OOP) principles to structure the library, utilizing robust constructors and internal scaling logic for accurate floating-point precision.

**RISC-V Pipelined Processor Simulator** | C++ | Course Project

- Developed a 5-stage pipelined RISC-V processor simulator by extending a single-cycle implementation.
- Implemented key pipeline features including hazard handling, branch prediction, and cache simulation to improve execution efficiency.
- Added configurable memory hierarchy components and evaluated the impact of different architectural design choices.

## POSITIONS OF RESPONSIBILITY

**Events Coordinator | Diesta, IITH :**

- Assisted in organization of Diesta '25, the inter-departmental sports and cultural fest of IITH.
- Managed coordination between all the parties concerned with the several sports and cultural events and co-charted the venue-time and equipment allocation.

**Multimedia Coordinator | Alumni Cell, IITH:**

- Assisted in management of real-time audio-visual operations and content capture for a tenure's worth of events related to the Alumni Cell like, Graduation day, Alumni day, etc.
- Coordinated in the moment multimedia coverage for multiple Alumni networking events, handling live photography and videography.

## EXTRACURRICULAR

- Intermediate Gamedev - Learning Game Development in **Godot** - an open source game engine for 2D and 3D games and XR experiences.
- Intermediate 3D Artist - Intermediate proficiency in 3D modeling, rendering, texturing and scene composition with an accumulated experience of 4+ years.